

EMBEDDED HACKING

Reverse Engineering the Raspberry Pi Pico 2

Operation Ghost Light: Classified Intelligence Briefing

ARM Cortex-M33 • GDB • Ghidra

October 9, 2026

For educational and defensive security research only

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|                TOP SECRET // NOFORN                |
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| OPERATION GHOST LIGHT                               |
| CLASSIFIED BRIEFING: LIVE CTF 0x05                  |
| NATIONAL SECURITY AGENCY / GMU RHET                 |
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1. The Intercept

Adversarial forces have deployed ground-based C-UAS laser dazzlers that track and blind friendly optical drones. The dazzlers are dynamically configured in the field using an infrared (IR) remote control running a customized NEC protocol.

2. The Mandate

We have captured an intact dazzler control board. The firmware uses deeply nested, dynamic conditional logic (`if / else if / switch`) to parse the raw hex codes from the IR receiver. There is a "Friendly Fire Lockout" code buried deep in the dynamic conditionals that permanently disables the dazzler.

You must step through the dynamic branches in GDB, trace the `CMP` and `BNE` chains, and reverse engineer the exact 32-bit NEC hex code required to trigger the lockout state.